

Acute knee sepsis after intramedullary nailing of open tibial shaft fractures: a comparison of intra-articular and extra-articular approaches

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Abstract

Objectives: To compare the incidence of acute knee sepsis after intramedullary nailing of open tibial shaft fractures via intra-articular and extra-articular approaches.

Design: Retrospective cohort analysis.

Setting: Two academic Level I trauma centers.

Patients/Participants: Adults treated with intramedullary nailing for open tibial shaft fractures between 2010 and 2023 with a minimum of 3-month follow-up.

Intervention: Intramedullary nailing using an intra-articular (suprapatellar) or extra-articular (medial parapatellar, transtendinous, or lateral parapatellar) approach.

Main Outcome Measurements: Acute knee sepsis, deep/superficial infection, reoperation, and nonunion.

Results: A total of 293 patients with 295 fractures met inclusion criteria (75% male, median age 41 years, 28% active smokers). Ninety fractures underwent intra-articular nailing, and 205 underwent extra-articular nailing. There were no differences in injury energy, Gustilo–Anderson classification, or rates of external fixation between groups. Acute knee sepsis occurred in one patient (1.1%) in the intra-articular group and one patient (0.5%) in the extra-articular group. Deep infection occurred in 8.9% for intra-articular and 9.3% for extra-articular ($P > 0.999$), and aseptic nonunion in 8% vs 7%, respectively ($P > 0.999$).

Conclusions: Acute knee sepsis after intramedullary nailing of open tibial shaft fractures was rare (0.7%). Knee sepsis was more common after suprapatellar nailing than extra-articular approaches (1.1% vs 0.5%), but the results are not statistically significant, and the study lacks power to draw definitive conclusions about relative risk with such low base rates. These findings may assist surgeons with approach selection and patient counseling, taking into account injury characteristics and technical considerations related to the procedure that are case-specific.

Keywords: open tibia fracture, tibial nail, knee sepsis, suprapatellar, infrapatellar, lateral parapatellar

Introduction

Tibial shaft fractures occur at an incidence of 16.9 per 100,000 annually, with up to 1 in 4 presenting as an open fracture.^{1,2} Open tibial shaft fractures are associated with infection in 18% to 30%

of cases, depending on the severity of the injury.^{2–4} More severe Gustilo–Anderson fracture types, higher Body Mass Index (BMI), increased time to soft tissue closure, and need for flap coverage are associated with increased risk of developing infection.^{5,6}

R. K. Wagner received support from the Marti-Keuning Eckhardt Foundation (Amsterdam, the Netherlands), the Cultuurfonds (Amsterdam, the Netherlands), the VSB Foundation (Utrecht, the Netherlands), and The Prof. Michiel-van Vloten Foundation (the Hague, the Netherlands). P. Grisdela Jr., has received an AO Trauma North America Fellow's research grant for \$10,000 for work not related to this project. In addition, P. Grisdela Jr., has received support from the AO Trauma North America for the fellow's experience in Davos, Switzerland (course registration, travel and food) and from Stryker for travel and lodging for courses. A. Aneja has received honoraria from AO Trauma North America for education, receives research support from the Orthopaedic Trauma Association, Department of Defense, and the American Orthopaedic Foot and Ankle Society, and is a committee member for the Orthopaedic Trauma Association and the American Academy of Orthopaedic Surgeons. T. V. Ly is on the speaker's bureau for AO Foundation. M. J. Weaver receives royalties from OsteoCentric for device design unrelated to this project. The remaining authors have no relevant financial or nonfinancial interests to disclose.

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There was no source of funding for this investigation.

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OTAI (2026) e516

Received: 9 April 2025 / Accepted: 10 July 2026

Published online 11 September 2026

<http://dx.doi.org/10.1097/OI9.0000000000000516>

Intramedullary nailing is a common treatment for open tibial shaft fractures and can be performed through a suprapatellar, infrapatellar, or lateral parapatellar approach.^{7,8} The suprapatellar portal passes through the patellofemoral joint, which carries a theoretical risk of septic arthritis in the setting of open fracture. However, retrograde femoral nailing uses a similar patellofemoral entry portal, and prior studies have demonstrated the relative safety of this procedure for the treatment of open fractures.^{9–11} Medial parapatellar, transtendinous, and lateral parapatellar approaches for tibial nailing are sometimes considered to be extra-articular and, therefore, to carry lower risk of septic arthritis in the setting of open fracture.^{8,12,13}

Studies reporting on knee sepsis after intramedullary nailing of open tibial shaft fractures are limited. One prior comparative study reported no significant difference in acute knee sepsis between the suprapatellar and infrapatellar approaches but did not evaluate the risk of knee sepsis associated with the lateral parapatellar approach.^{12,13} The primary aim of the present study was to compare rates of acute knee sepsis and complications for open tibial shaft fractures treated with intra-articular versus extra-articular intramedullary nailing techniques. Our null hypothesis was that there would be no difference in rates of acute knee sepsis between intra-articular (suprapatellar) versus extra-articular (medial parapatellar, transtendinous, or lateral parapatellar) approaches.

Methods

Patient Selection

IRB approval (protocol number: 2023P003322) was obtained, and the study was conducted in compliance with the ethical guidelines of the participating institutions. All consecutive adult patients treated with intramedullary nailing for open tibial shaft fractures at 2 Level I trauma centers between January 2010 and December 2023 were identified from an institutional database. Patients were identified using Current Procedural Terminology (CPT) codes 27752, 27750, 27759, 11010, 11011, and 11012. Patients with insufficient medical documentation, less than 3 months of follow-up, and pathological fractures were excluded. A minimum follow-up of 3 months was chosen because 60% to 80% of deep infections manifest within this time period.¹⁴ Patients undergoing nonoperative management, with fractures extending into the metaphysis or articular surface, periprosthetic fractures, and fractures in patients with prior plate fixation, were also excluded.

Data collection

Variables were collected from hospital electronic medical records by chart review. Demographic variables included age, gender, BMI, diabetes, and smoking status. Injury and fracture characteristics included injury energy (high-energy mechanisms were defined as motor vehicle accidents, falls from height, and high-velocity impacts, whereas low-energy mechanisms were defined as simple falls from standing height and twisting injuries during routine activities), OTA/AO classification, and Gustilo–Anderson classification.^{15,16} Treatment characteristics included the type of initial treatment (intramedullary nailing, external fixation, damage control plating), time to intramedullary nailing in temporized patients, surgical approach for intramedullary nailing (intra-articular or extra-articular), use of antibiotic cement spacer (i.e., Masquelet), fasciotomy, flap coverage, type of systemic antibiotics given at time of incision, and use of local

antibiotic powder during intramedullary nailing. For the purposes of this study, the intra-articular (suprapatellar) approach was performed through a standard trans-quadriceps tendon incision using a cannula designed to traverse the patellofemoral joint, with the knee in a semi-extended position. Extra-articular techniques were grouped together. These included the standard infrapatellar approach for tibial nailing with the knee flexed over a triangle using a limited medial parapatellar or transtendinous incision. The lateral parapatellar approach was also considered to be extra-articular and was performed using a small incision along the lateral border of the patella, with dissection and nail insertion remaining extracapsular with the knee in a semi-extended position.¹⁷ All fixation surgeries were performed by fellowship trained orthopaedic surgeons, with the choice of surgical approach and wound closure at each surgeon's discretion.

Outcome variables

The primary outcome was acute knee sepsis, defined as a positive culture from a knee aspiration or arthrotomy. Culture results from the joint fluid were compared with separately obtained canal cultures. Secondary outcomes included rates of deep infection, superficial infection, nonunion, and hardware removal. Deep infection was based on Fracture-Related Infection consensus criteria and defined as any infection requiring operative debridement, whereas superficial infection was defined as wound complications requiring oral antibiotics after definitive fixation.¹⁸ Nonunion and hardware removals were determined based on review of clinical documentation and surgeon notes. Nonunion was defined as a return to the operating room for revision fixation to address a nonhealing fracture.

Statistical analysis

Variables are presented as frequencies with percentages for categorical variables and as medians with interquartile range (IQR) for continuous variables. Comparisons between groups of interest were performed using Wilcoxon rank-sum test and Fisher exact test. A *P*-value of <0.05 was considered statistically significant. All statistical analyses were performed using R (Version 4.3.0.21, Vienna, Austria). The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Results

A total of 481 patients with open tibial shaft fractures treated with intramedullary nailing were identified, of which 293 were included (Fig. 1). Two patients had bilateral open tibial shaft fractures, resulting in 295 fractures being included in the study. Patients excluded from the final analysis were similar to the included cohort with respect to age (median 40 vs. 41 years), sex (73% vs 75% male), BMI (median 25.2 vs 27), and prevalence of diabetes (11% vs 7%), although the excluded group had a higher rate of current smoking (37% vs 28%). The median patient age was 41 years (IQR 27–56) and 75% were male (Table 1). The median BMI was 27 (IQR 24–32), 19 patients (7%) had diabetes, and 73 (28%) were current smokers. A total of 247 injuries (84%) resulted from high-energy mechanisms, most often from motor vehicle crashes (*n* = 120, 41%). Fractures were classified as OTA/AO 42A in 170 (58%), 42B in 62 (21%), and 42C in 63 (21%). Open fracture types were Gustilo–Anderson type I in 67 (23%), type II in 90 (31%), and type III in 135 (46%) cases (Table 1).

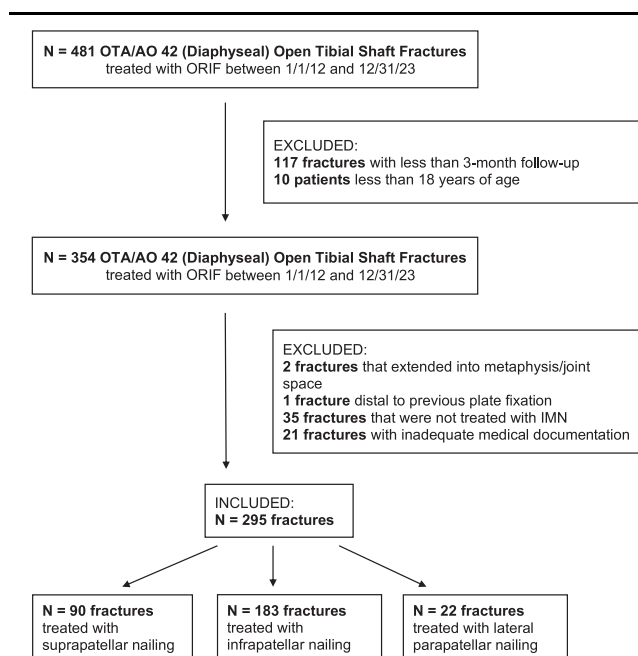


Figure 1. Flow diagram depicting patient selection.

Patients undergoing intra-articular approaches had lower BMI ($P = 0.002$), were more likely to be smokers ($P = 0.027$), and had more OTA/AO 42C injuries ($P = 0.029$) (Table 1).

Treatment characteristics are displayed in Table 2. Surgical approaches for intramedullary nailing were 90 intra-articular (31%) and 205 extra-articular (69%). Among extra-articular approaches, there were 183 infrapatellar (medial parapatellar or transtendinous) (89%) and 22 lateral parapatellar (11%). Twenty-seven (9%) fractures underwent temporary external fixation, and 4 (1%) underwent damage control plating before definitive treatment with intramedullary nailing, with a median

time to intramedullary nailing of 4.0 days (IQR 3.0–9.0). There was no significant difference in rates of provisional external fixation between the groups ($P = 0.058$). Eight (3%) fractures were treated using the Masquelet technique.

Primary outcomes

One patient (1.1%) in the intra-articular group and one patient (0.5%) in the extra-articular group developed acute knee sepsis after definitive intramedullary nailing confirmed by knee aspiration and intraoperative cultures of the medullary canal (Table 3). There was no statistically significant difference in the rate of acute knee sepsis between the intra- and extra-articular groups (odds ratio, 2.29; 95% confidence interval [CI], 0.14–37). A post-hoc sample size estimation based on observed acute knee sepsis rates indicated that approximately 3200 patients per group would be required to achieve 80% power at a two-sided α of 0.05, assuming equal distribution of patients between groups.

The patient with a septic knee in the intra-articular group was a 57-year-old male pedestrian struck by a motor vehicle with a Gustilo–Anderson type IIIA open tibial shaft fracture. An external fixator was placed at the time of initial irrigation and debridement and injury presentation. He was converted to intramedullary nailing using a suprapatellar approach on postinjury day 5. In the presence of the IM nail, one of the previous external fixator sites demonstrated persistent drainage, which was managed with local wound care and oral antibiotics. Approximately 2 months after the intramedullary nailing procedure, he presented with acute knee pain and effusion, which was confirmed as methicillin-susceptible *Staphylococcus aureus* septic arthritis by outpatient arthrocentesis. An exchange antibiotic nail was subsequently placed through an infrapatellar insertion approach. Intraoperatively, an irrigation and debridement of the knee was performed by a medial arthrotomy and was then closed before the exchange nailing. Cultures obtained separately from the knee joint and tibial reaming both yielded methicillin-susceptible *S. aureus*,

Table 1
Demographics.

	All (n = 295)	Intra-articular (n = 90)	Extra-articular (n = 205)	P
Age, median (IQR)	41 (27–56)	37 (27–54)	42 (27–57)	0.558
Male, n (%)	221 (75%)	67 (74%)	154 (75%)	0.885
BMI, median (IQR)	27 (24–32)	26 (23–30)	28 (25–33)	0.002*
Diabetes, n (%)	19 (7%)	4 (4%)	15 (7%)	0.446
Current smoker, n (%)	73 (28%)	31 (37%)	42 (23%)	0.027*
Injury mechanism, n (%)				0.017*
Motorized	120 (41%)	35 (39%)	85 (41%)	
Fall	78 (26%)	21 (23%)	57 (28%)	
Pedestrian struck	52 (18%)	16 (18%)	36 (18%)	
Sports	20 (7%)	13 (14%)	7 (3%)	
Other	25 (9%)	5 (6%)	20 (10%)	
Injury energy, n (%)				0.612
High	247 (84%)	77 (86%)	170 (83%)	
Low	48 (16%)	13 (14%)	35 (17%)	
OTA/AO classification, n (%)				0.029*
A	170 (58%)	45 (50%)	125 (61%)	
B	62 (21%)	17 (19%)	45 (22%)	
C	63 (21%)	28 (31%)	35 (17%)	
Gustilo–Anderson type, n (%)				0.764
Type I	67 (23%)	18 (20%)	49 (24%)	
Type II	90 (31%)	29 (33%)	61 (30%)	
Type III	135 (46%)	41 (47%)	94 (46%)	

Missing values for BMI (n = 72), ASA (n = 55), current smoker (n = 31), diabetes (n = 1), injury mechanism Gustilo–Anderson type (n = 3).

* $P < 0.05$.

Table 2**Treatment characteristics.**

	All (n = 295)	Intra-articular (n = 90)	Extra-articular (n = 205)	P
Initial treatment, n (%)				0.058
IMN	264 (89%)	76 (84%)	188 (92%)	
External fixation	27 (9%)	11 (12%)	16 (8%)	
Damage control plating	4 (1%)	3 (3%)	1 (1%)	
Time to IMN in temporized patients, median (IQR)	4.0 (3.0–9.0)	4.0 (2.0–9.0)	5.0 (4.0–8.0)	0.575
Masquelet, n (%)	8 (3%)	4 (4%)	4 (2%)	0.253
Fasciotomy, n (%)				
Therapeutic	19 (6%)	6 (7%)	13 (6%)	>0.999
Prophylactic	18 (6%)	8 (9%)	10 (5%)	0.194
Wound closure, n (%)				
Primary closure	224 (76%)	70 (78%)	154 (75%)	0.66
Skin graft	52 (18%)	14 (16%)	38 (19%)	0.62
Flap	56 (19%)	18 (20%)	38 (19%)	0.75
Type of antibiotics given at time of incision, n (%)				0.008*
Cefazolin	179 (69%)	57 (64%)	123 (72%)	
Cefazolin + vancomycin	40 (15%)	20 (22%)	20 (12%)	
Clindamycin	9 (4%)	2 (2%)	7 (4%)	
Vancomycin	8 (3%)	6 (7%)	2 (1%)	
Other	22 (9%)	4 (5%)	18 (11%)	
Vancomycin powder, n (%)	9 (3%)	6 (7%)	3 (2%)	0.026*
Follow-up weeks, median (IQR)	41 (24–64)	42 (26–59)	41 (23–66)	0.94

Missing values for type of antibiotics given at time of incision (n = 37).

* $P < 0.05$.

IMN, intramedullary nailing.

matching the knee aspiration cultures obtained in clinic. There was cortical bridging on imaging 10 months after nail exchange.

The patient with a septic knee in the extra-articular group was a 53-year-old man who was involved in a motorcycle accident who presented with a Gustilo–Anderson type IIIB open tibial shaft fracture complicated by compartment syndrome requiring acute four-compartment fasciotomy. He underwent acute intramedullary nailing via an infrapatellar approach at the time of injury and soleus rotational flap with split-thickness skin grafting by plastic surgery 3 days after definitive nailing. Two months after the intramedullary nailing procedure, he had a draining wound over the distal leg adjacent to the flap and also complained of knee pain. Cultures of the sinus tract at the flap demonstrated methicillin-susceptible *S. aureus* and matched cultures taken from the knee joint. He was treated with surgical debridement and irrigation of the knee and local wound care for the flap. The nail was retained and infection suppressed until he achieved union and then underwent hardware removal 12 months after his knee irrigation and debridement.

Secondary outcomes

The intra-articular group had 8 (8.9%) deep infections and 4 (4.4%) superficial infections. The extra-articular group had 19 (9.3%) deep infections and 11 (5.4%) superficial infections. There were no significant differences in the rate of deep infection or superficial infection between the intra-articular and extra-articular groups (Table 3). No significant differences in rates of

aseptic nonunion ($P > 0.999$), infected nonunion ($P = 0.226$), or hardware removal ($P = 0.825$) were observed between the groups.

Discussion

Open tibial shaft fractures are commonly treated with intramedullary nailing, which can be performed via one of 3 approaches: suprapatellar, infrapatellar, or lateral parapatellar.^{7,8} Suprapatellar nailing uses an entry portal that traverses the patellofemoral joint, raising concerns about potentially introducing infection into the knee joint. The infrapatellar and lateral parapatellar approaches are often considered extra-articular, although cadaver studies have suggested that violation of the joint capsule may occur with both of these approaches.^{8,12,13} Previous clinical studies have found low rates of acute knee sepsis after suprapatellar nailing of open tibial shaft fractures with no cases reported after infrapatellar nailing.^{12,13} The current study demonstrates a clinical risk with both intra-articular and extra-articular approaches, but the prevalence was higher in the intra-articular group. This information may aid surgeons in patient counseling, and surgeons should maintain familiarity with both approaches to tailor treatment to the specific patient and clinical scenario.

In the present study, the overall rate of acute knee sepsis was 0.7% with only 2 patients developing acute knee sepsis after intramedullary nailing (one suprapatellar and one infrapatellar approach). The rate was higher in the suprapatellar group

Table 3**Infection characteristics.**

	All (n = 295)	Intra-articular (n = 90)	Extra-articular (n = 205)	P
Knee sepsis, n (%)	2 (0.7%)	1 (0.5%)	1 (1.1%)	0.518
Deep infections, n (%)	27 (9.2%)	8 (8.9%)	19 (9.3%)	>0.999
Superficial infections, n (%)	15 (5.1%)	4 (4.4%)	11 (5.4%)	>0.999

compared with the infrapatellar group (1.1% vs 0.5%). Mitchell et al previously reported no cases of knee sepsis after suprapatellar nailing in a series of 139 open tibial shaft fractures, but follow-up was limited to 30 days. Both cases of knee sepsis in the present study were diagnosed more than 30 days postoperatively. In the only prior comparative study, Marecek et al reported 2 cases of knee sepsis after intramedullary nailing of 289 open tibial shaft fractures (147 suprapatellar and 142 infrapatellar), both in patients with a suprapatellar approach.^{12,13} They concluded that although there was no statistically significant difference in rates of knee sepsis between suprapatellar and infrapatellar approaches, the risk of knee sepsis after suprapatellar nailing is low but present. Finally, there are 2 prior case reports of knee sepsis after intramedullary nailing with the suprapatellar approach, but none associated with the infrapatellar approach.^{19,20}

In this multicenter comparative study of intra-articular vs extra-articular approaches, acute knee sepsis was rare after intramedullary nailing of open tibial shaft fractures. The results are consistent with prior studies demonstrating that the use of an intra-articular technique is generally safe with a low risk of acute knee sepsis. Contrary to prior studies and consistent with cadaver studies demonstrating intra-articular damage after infrapatellar and lateral parapatellar nailing, this study also demonstrated a low but nonzero risk of acute knee sepsis with an extra-articular technique. The rate of acute knee sepsis in this cohort after intramedullary nailing using an intra-articular technique (1.1%) was twice the rate after use of an extra-articular technique (0.5%), but the study was underpowered to detect a difference between the 2 groups. Surgeons using either intra-articular or extra-articular approaches for nailing open tibial fractures should be aware of the low but nonzero risk of acute knee sepsis associated with both techniques. Although the overall incidence was low, the risk of knee sepsis may be higher after suprapatellar nailing than with extra-articular approaches. Larger, prospective studies are needed to define the relative risk introduced by the choice of one approach over the other. This information may inform surgeons' choice of approach and be useful for counseling patients.

Suprapatellar nailing via an intra-articular approach may offer distinct advantages in certain clinical scenarios, including facilitation of fracture reduction and maintenance of alignment.^{21,22} When treating open tibial shaft fractures, these potential benefits should be weighed alongside the possibility of a small increase in the risk of acute knee sepsis associated with the intra-articular approach. Furthermore, given the potential infection-related concerns associated with traversing the knee joint in open injuries, proficiency with extra-articular techniques, including infrapatellar nailing, remains an important skill for trainees and early-career surgeons. Familiarity with both approaches allows surgeons to tailor treatment to the specific clinical scenario, fracture characteristics, and patient risk profiles.

This study has several limitations that are inherent to the retrospective design. The relatively short follow-up period of 3 months could have resulted in missing patients who experienced complications after this timeframe or sought follow-up care at other institutions. However, this timeframe is consistent with the other prior comparative study by Marecek et al and considered adequate to capture most cases of acute knee sepsis. The limited number of patients in each group and low rate of acute knee sepsis overall means the study is underpowered to detect a significant difference in risk of acute knee sepsis between groups or provide useful information about

relative risk between approaches. In addition, the findings of acute knee sepsis are consistent with prior studies and offer clinical evidence to corroborate the theoretical risk of contamination even with an extra-articular approach. Finally, risk of knee sepsis may be higher for more severe open fracture types or grossly contaminated fractures. Subgroup analysis for such an effect was limited by sample size in this study, but surgeon discretion is paramount to avoid introduction of gross contamination through any nailing portal.

Conclusions

Acute knee sepsis after intramedullary nailing of open tibial shaft fractures was very rare with an overall rate of 0.7%. Knee sepsis was more common after suprapatellar nailing compared with extra-articular approaches (1.1% vs 0.5%), but the results are not statistically significant, and the study lacks power to draw definitive conclusions about relative risk with such low base rates. These findings may assist surgeons with choice of approach and patient counseling, taking into account injury characteristics and technical considerations related to the procedure that are case-specific.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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